DSI Showcase – DS Team 2
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UCI Bank Marketing:

Uncovering the Key Drivers of Term Deposit Subscriptions

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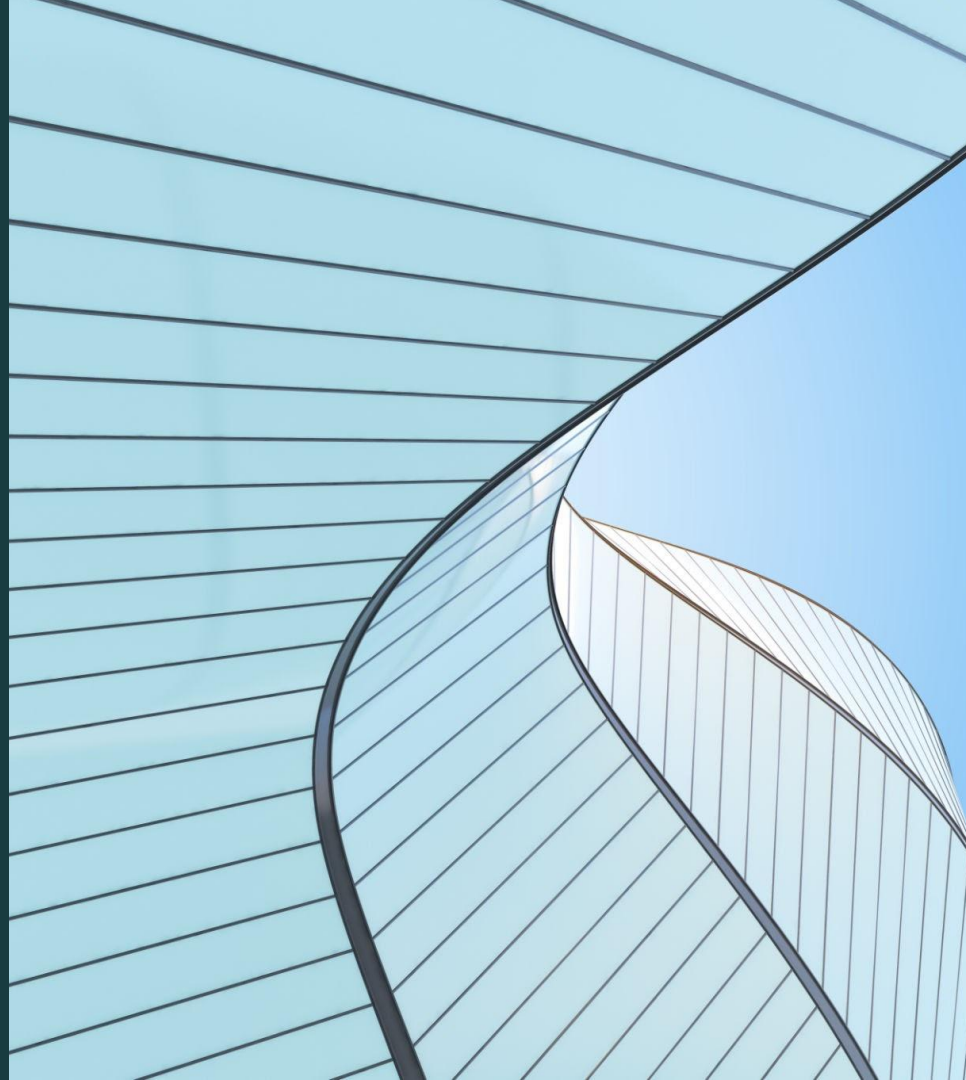
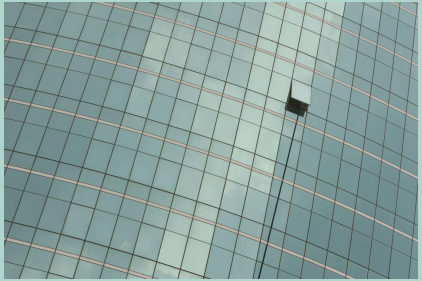




THE PROBLEM

UCI Bank often rely on mass-calling strategies for term deposit campaigns. This results in:

- Low conversion rates
- High operational costs
- Agent time spent on low-probability customers
- Customer fatigue from excessive follow-ups

The gap: There is no optimized targeting strategy based on customer history, financial profile, and campaign behavior.

OBJECTIVE

Using exploratory data analysis and machine learning models, we optimize bank marketing campaigns by targeting the right customers to increase term deposit subscription rates and reduce wasted outreach.

Questions:

- Which client segments are most likely to subscribe to a term deposit?
- What key factors drive a client's decision to subscribe?
- Can we predict which clients will subscribe?
- How can banks use these insights to improve campaign performance?

DATASET OVERVIEW: VARIABLES

The dataset comes from the UCI Bank Marketing dataset contains multi-year campaign data; approximately 45,000 structured customer records from direct marketing campaigns, *with a binary target variable y indicating whether a client subscribed to a term deposit (yes/no).*

Client Data

Basic customer background characteristics used for segmentation:

- → age
- → job
- → marital status
- → education

Financial Data

Customer financial condition and credit obligations:

- → balance (account balance)
- → housing (loan status)
- → loan (personal loan status)
- → default (credit in default)

Campaign Data

Information related to the current marketing campaign contact:

- → contact (cell or tele)
- → last contact month
- → call duration in secs.
- → number of contacts during current campaign

Previous Campaign Data

Customer interaction history from prior campaigns:

- → pdays (days since last contact)
- → previous (number of previous contacts)
- → poutcome (outcome of previous campaign)

DATASET OVERVIEW: RISKS, UNKNOWNS & BIASES

	1	2	3	4
RISKS	**Data Leakage:** Call duration strongly predicts subscription but is only known after the call. Using it in a predictive model for pre-call targeting would lead to unrealistic predictions.	**Overfitting to Historical Campaign Behavior:** Patterns identified in the dataset may reflect past campaign strategies rather than true customer preferences.	**Operational Misinterpretation:** Insights such as longer call duration increasing success could be misinterpreted as a strategy rather than a result of customer interest.	**Limited Generalizability:** The dataset comes from one Portuguese bank and may not represent customer behavior in other institutions or markets.
UNKNOWNS	**Customer Income and Wealth:** The dataset does not include income or net worth, which likely influence subscription decisions.	**Customer Relationship Value:** Information such as customer tenure, savings history, or overall relationship with the bank is not available.	**Marketing Message Content:** The dataset does not capture what was said during calls or the quality of the sales interaction.	**External Economic Conditions:** Interest rates, inflation, and economic conditions at the time of the campaign are not included but may affect customer decisions.
BIASES	**Historical Targeting Bias:** Certain customer groups may have been contacted more frequently in past campaigns, influencing observed outcomes.	**Selection Bias:** The dataset only includes customers who were contacted, not the entire customer base.	**Survivorship Bias:** Customers who stayed with the bank long enough to be contacted may differ from those who left earlier.	**Response Bias:** Customers who answer calls may systematically differ from those who do not answer marketing calls.

OUR PROCESS

Applying structured analysis to transform raw customer data into targeted marketing strategy.

01
DATA CLEANING

Prepared and structured the dataset for analysis.

- Checked for missing or inconsistent values
- Standardized categorical variables
- Verified data types and distributions
- Removed or reviewed extreme outliers

02
EDA

Identified patterns and relationships within the data.

- Analyzed subscription rates across segments
- Compared demographic and financial groups
- Evaluated balance distributions
- Reviewed campaign variable behavior

03
ANALYSIS

Measured the impact of key variables on subscription outcomes.

- Compared conversion rates by previous campaign outcome
- Evaluated effect of call duration and contact frequency
- Assessed impact of financial stability (balance, loans, default)
- Analyzed performance by campaign month

04
BUSINESS IMPACT

Focused on actionable business insights rather than model complexity.

- Identified high-probability customer segments
- Determined optimal campaign timing
- Recommended efficient follow-up strategy
- Prioritized financially stable customer profiles

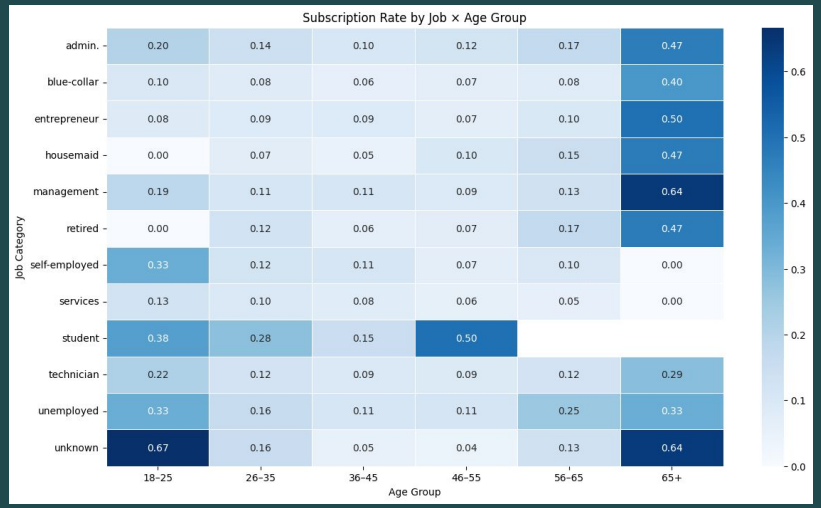
TOOLS LEVERAGED

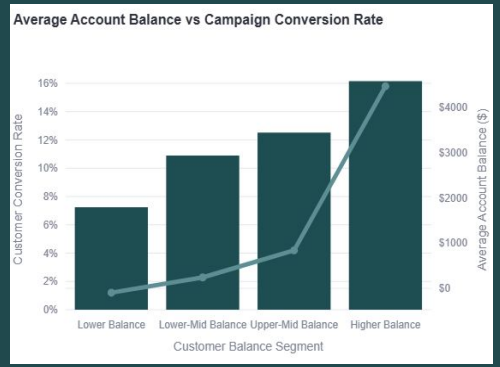
Python	Data cleaning, analysis, and exploration
Pandas & NumPy	Data manipulation and preprocessing
Matplotlib / Seaborn	Data visualization and pattern identification
Streamlit	Interactive data presentation and deployment
Jupyter Notebook	Development and analysis environment

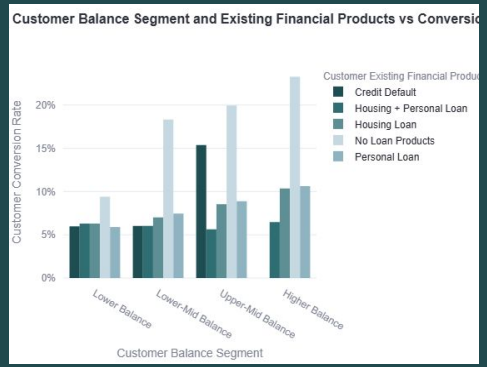
KEY CUSTOMER INSIGHTS

- Customers at life-stage transition points—such as **retirees, students, and some unemployed individuals** - are more likely to subscribe, while *working-age customers (26–55) in operational roles* show lower responsiveness. Additionally, **higher education levels** are associated with a slightly greater likelihood of subscription.

- Customers with **strong financial positions** - *high account balances and minimal existing debt* - are the most responsive to marketing campaigns, making them the most effective target segment for future outreach.

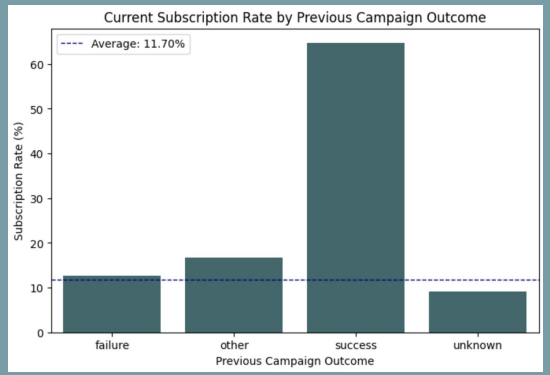


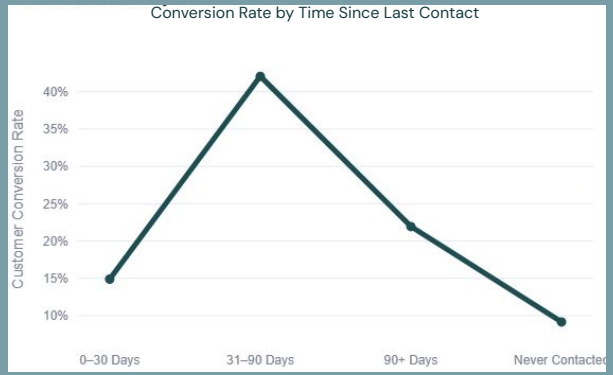


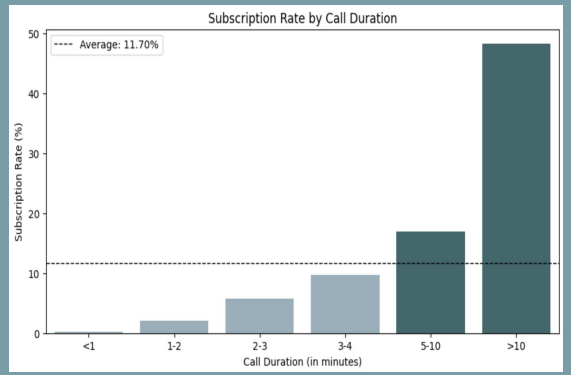


KEY INSIGHTS FROM PREVIOUS CAMPAIGNS

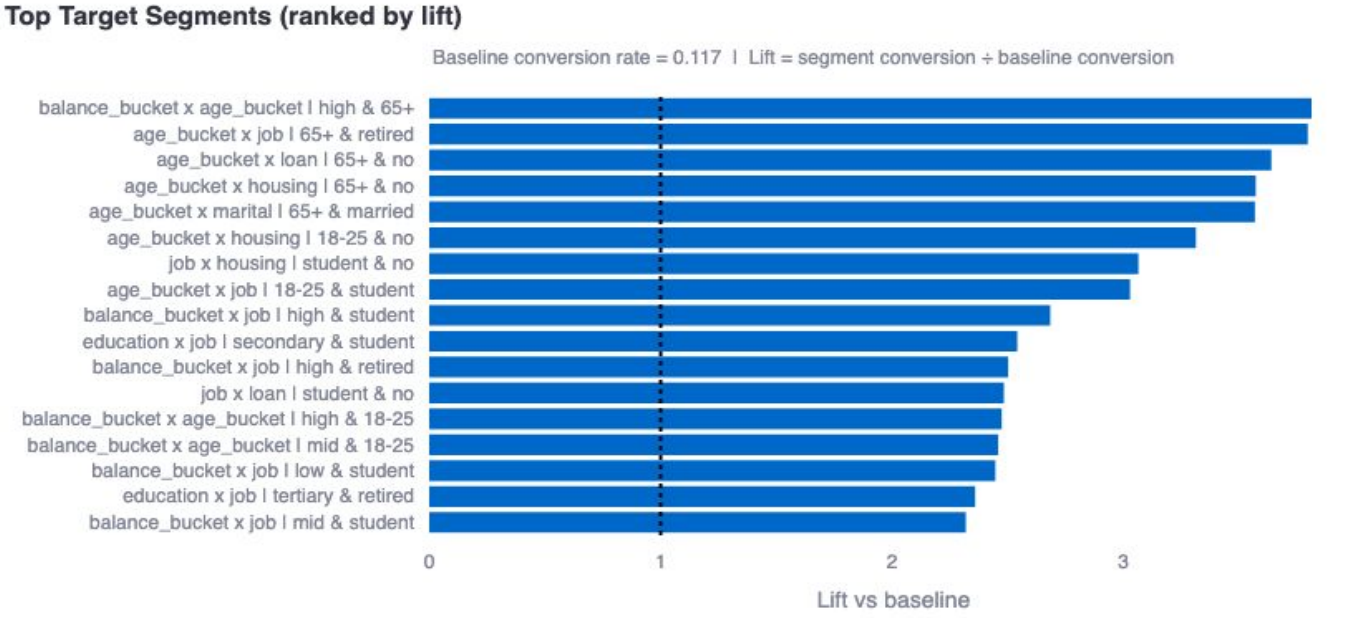
- **Customers who previously subscribed are the most promising targets**, consistently showing the highest likelihood of converting again in future campaigns.
- **Customer engagement factors matter**, with recently contacted customers and longer call durations associated with higher subscription rates.
- **Campaign timing influences results**, with peak subscription activity occurring in March and during the later months of the year.
- **Over-contacting customers reduces effectiveness**, as repeatedly targeting previously unsuccessful customers leads to diminishing returns.







KEY SEGMENTATION INSIGHTS



MODEL METHODS

Model Performance Summary

Model	Precision	Recall	AUC	Business Impact (per 100 calls)
KNN (k=51)	77.67%	6.05%	0.7610	78 subscribers, 22 wasted
Logistic Regression	53.57%	1.13%	0.7322	54 subscribers, 46 wasted
Random Forest	**64.53%**	**21.33%**	**0.7905**	**65 subscribers, 35 wasted**

The table shows that Random Forest provides the best balance between identifying potential subscribers and minimizing wasted calls, making it the most useful model for improving campaign targeting.

Streamlit Dashboard

BUSINESS STRATEGY

FUTURE WORK

The following opportunities represent areas that were beyond the scope of this project but could further strengthen insights and business impact for the bank:

Advanced Predictive Modelling

Explore additional modeling techniques to more accurately predict customer conversion and identify the strongest drivers of subscription.

Incorporate Additional Data Sources

Integrating more customer-related datasets—such as transaction history, income estimates, digital engagement data, or customer tenure— for a more complete view of customer financial behavior.

Deeper Analysis of Business Impact

Further analysis could quantify the business impact of the identified insights even more!

Thank you